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DSMC SIMULATIONS OF SQUEEZE FILM UNDER A HARMONICALLY OSCILLATING MICRO RF SWITCH WITH LARGE TIP DISPLACEMENTS

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ABSTRACT

The two-dimensional unsteady behavior of a rarefied gas film under an oscillating micro-cantilever RF switch is presented. The microbeam, undergoing a parabolic deflection profile, is allowed to oscillate harmonically between its equilibrium position and the fixed substrate underneath for large beam-tip displacements. The gas film dynamics in terms of the flow field velocity and fluid forces exerted on the oscillating microbeam are discussed. The numerical technique used to model the rarefied gas flow is the Direct Simulation Monte Carlo (DSMC) method where the Knudsen (Kn) number is greater than 0.01 (ie. non-continuum regime). Unlike previous work in literature, the beam undergoes large deflections, which requires implementation, in DSMC, of a more realistic molecule-beam reflection behavior based on the instantaneous beam's position and velocity. The effects of inertia, both local acceleration (St) and convection term (Re), and compressibility (Ma) on the gas film dynamics are examined over ranges of oscillating frequencies, velocity amplitudes, and microbeam's lengths.

NOMENCLATURE

DSMC Direct Simulation Monte Carlo.
 L Microbeam length.
 V_a Microbeam tip velocity amplitude.
 f Oscillation frequency.

M_d Molecular diameter.
 M_m Molecular mass.
 γ Specific heat ratio.
 ρ Density.
 λ Mean free path.
 a Speed of sound.
 μ Dynamic viscosity.
 k Boltzmann constant.
 h_0 Gap film thickness.
 Kn Knudsen number.
 Re Reynolds number.
 St Strouhal number.
 Ma Mach number.
 δ Microbeam tip displacement amplitude.
 y_{tip} Microbeam tip displacement.
 $\dot{y}_{tip}$ Microbeam tip velocity.
 θ Microbeam tip deflection angle.
 τ_{conv} Convection time scale.
 τ_{diff} Diffusion time scale.
 τ_{osc} Oscillation time scale.
 τ_{coll} Molecular collision time scale.
 T_0 Ambient temperature.
 P_0 Ambient pressure.

INTRODUCTION

With the development of MEMS technology, the interaction between a moving microstructure and the surrounding fluid gained attention due to the impact on the dynamics of motion of

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the microstructure. An example is an RF MEMS switch. To better understand the behavior of an RF MEMS switch, many models have been suggested in literature to describe the physical behavior of the gas (usually air) film between the vibrating plate or beam and the underneath fixed substrate. With the emerging micro technology in the early 90's, many researchers have reflected back on Reynolds lubrication theorem, which was first derived by Blech [1], due to the interest in fluid film behavior in bearings. Bao et al. [2] proposed a new model for air damping coefficients in vibratory micro-sensors operating under low ambient pressure. Their model uses energy transfer between the moving molecules and the vibrating plate and therefore surpasses Christian's model [3] in two main aspects: the new model is a function of the structure geometry and takes into account the presence of boundaries near the vibrating structure. The effect of the presence of a fixed boundary near a vibrating substrate was discussed by Newell [4] who studied its effect on the damping coefficient of the squeezed gas film (i.e. quality factor). Hutcherson and Ye [5] modeled the quality factor of a vibrating microbeam and discussed the validity of previous models in literature depending on the boundary conditions assumed (i.e. free oscillating structure as compared to a structure oscillating near a fixed wall). Li and Fang [6] extended the work of Hutcherson and Ye that was limited to parallel plate vibrations to account for rotating plates and bending of microbeams. Veijola et al. [7] used analytical models for the gas flow damping coefficient to solve for the capacitance of a perforated plate RF MEMS switch undergoing small amplitude vibrations using circuit analysis. The validity of their model is restricted to relatively low switching frequencies (and consequently Reynolds number) and inertia free flows. Veijola [8] extended the Reynolds squeezed film damping equation model by introducing the gas rarefaction effects and inertia effects (limited to the local acceleration term) and used reduced order modeling (circuit analysis) to solve the resulting improved equations. Veijola and Lehtovuori [9] studied the effect of microstructures oscillating at very high frequencies for small vibration amplitudes and showed that the squeezed gas gets trapped between the oscillating and fixed structures. Gallis and Torczynski [10] modified the Reynolds equation model for squeezed film damping by correcting for the pressure boundary conditions at the beam tips compared to the trivial ambient pressure conditions used before. They used the NavierStokes slip-jump (NSJJ) and the Direct Simulation Monte Carlo (DSMC) methods to determine the new pressure boundary conditions coefficients. J W Lee et al. [11] extended the work of [8] and [10] by accounting for the higher order bending modes in deriving a new squeezed film damping ratio model. Such an extension provided a improved model that gave better results as compared to experimental data carried by the authors themselves since it accounts for the bending of the beam structure. A main restriction in their work is that the cantilever beam deflection was restricted to less than two orders of magnitude smaller than the film gap height to ensure the

validity of the linear damping model. Guo and Alexeenko [12] used the ellipsoidal statistical Bhatnagar-Gross-Krook model of the equilibrium distribution function in the quasi-steady Boltzmann kinetic model to derive expressions for the gas film damping coefficient and the quality factor. Their results showed the importance of the vibrating beam's structural damping as compared to the fluid damping in the vicinity of high Kn number (i.e. free molecular regime). However, their work was simplified by assuming a small vibration amplitude as compared to the film gap width and the cantilever beam length to be much larger than its width transforming the problem into a two-dimensional flow along the beam's width. Sumali et al. [13] studied the air drag force (i.e. free space damping) on a vibrating micro cantilever beam that is away from any fixed substrate. They conducted experiments to compare the gas damping coefficients with models of literature (ex. [3] [4] [14]) and verified that [14] was the most accurate in the free molecular and viscous regimes. Hosaka et al. [15] modeled the behavior of micro-cantilever beams by including damping in the squeeze film and on the free-space side of the beam, in addition to internal structural damping, and the support loss at the micro beam's base. They concluded that the squeeze and free space damping dominates the structural and support loss damping effects for small cantilever beams. As the beam's length increases, the structural and support loss damping effects are unchanged while the squeeze damping decreases at a higher rate than the free space damping. It is worth noting that the free space force was modeled based on a series of assumptions from Kokubun's bead model [16]. In their model, Kokubun et al. [16] proposed to replace the force model of a harmonically oscillating structure (beam or tuning fork) with a string of spheres since the equation of the drag force acting on a sphere can be easily derived. Li et al. [17] presented analytical models for the squeeze film damping films by solving the compressible Reynolds equation along with the beam deflection (i.e. Euler equation) under electrostatic actuation. The main contribution of their work is including the effect of static deflection on the beam caused by the steady state deflection for a step actuation voltage and the bias voltage for a harmonic driving actuation voltage. The analytical models were derived for two types of beams: fixed-fixed and fixed-free (cantilever) microbeams. Nayfeh et al. [18] modeled a fixed-fixed cantilever beam under the effect of large dc electrostatic driving force. The main advantage of this work was to simplify the clamped-clamped microbeam model into one structural finite-element solver rather than solving for the beam deflection, electrostatic actuation, and squeeze film behavior as three coupled nonlinear solvers.

METHODOLOGY

All the works in the literature just discussed have at least one of the following assumptions: 1) Microbeam vibrating at low speeds (i.e. small Re) 2) small vibration amplitude as compared

to film gap width (ie. linear damping models) 3) microbeam's length is much larger than the width and thickness (ie. reducing problem to two dimensional flow) 4) trivial boundary conditions are applied at beam's borders.

The objective of this paper is to develop the previous work of Diab and Lakkis [19] for a rigid oscillating parallel plate and use the Direct Simulation Monte Carlo (DSMC) ([20], [21], and [22]) method to model the behavior of the gas film between a 2D micro-cantilever beam (RF switch), whose tip undergoes large amplitude oscillations, and the fixed substrate underneath. The basic steps involved for the DSMC algorithm implemented in this work to model the non-continuum gas flow behavior around a 2D RF micro switch is presented in [19]. The cells were sized at about one-half the molecular mean free path and the simulation time step (0.1 nsec) was chosen to be less than the molecular collision time step to guarantee that, on average, a molecule spends about 2 to 3 time steps within a cell before moving into a neighboring cell. Moreover, every cell has been divided into four subcells where molecule collision pairs are chosen within one subcell and the thermodynamic properties are sampled at individual cells. All simulations have been performed for a number of periods to overcome quiescence effects. Figure 1 shows the two dimensional computational domain (ABCD) as modeled in the DSMC code where the microbeam is defined by the rectangle EFGH. AB and BC along with the microbeam faces (EF, FG, and GH) are all modeled as diffusive walls at ambient temperature. The enclosed region ADCC'IA'A contains the reservoir cells that are modeled to impose the desired boundary conditions of ambient temperature and pressure using the Maxwellian reservoir method. The reservoir cells are maintained at the desired boundary conditions by correcting for the number of molecules and their corresponding distribution (position and speed) at every time step to ensure that they are in thermodynamic equilibrium with their neighboring simulation cells. Unlike previous works in the literature that are limited to small vibration amplitudes or frequency ranges where the force exerted by the gas on the beam is modeled without detailed investigation of the gas behavior, in this work, the micro-cantilever beam is allowed to move at any desired velocity amplitude and frequency (i.e. the distance covered by the vibrating beam tip ranges from 1% to 50% of the gap film thickness) and the gas behavior is modeled by the interaction of molecules among themselves and the oscillating microbeam. The micro-cantilever beam's deflection at any distance x along the microbeam is given by the following parabolic profile

$$y(x,t) = \left(\frac{x}{L}\right)^2 y_{tip}(t) \quad (1)$$

where L is the length of the microbeam, and the tip is assumed to undergo a sinusoidal displacement as

$$y_{tip}(t) = \frac{\delta}{2} - \frac{\delta}{2} \cos(2\pi ft) \quad (2)$$

where δ and f are the tip displacement's amplitude and frequency.

Figure 2 shows the deflected microbeam (fixed end located at point O) at two consecutive time steps t_{i-1} and t_i . The DSMC algorithm checks for molecule/microbeam collisions for a molecule moving from its initial position (point I_0 at time t_{i-1}) to its new position (point I_f at time t_i) by linearly discretizing the beam's parabolic deflection profile into n intermediate curves. It is worth noting that as the number of interpolated curves (n) increases, the DSMC algorithm becomes more accurate in terms of capturing the point of impact between a moving molecule and the vibrating microbeam, but at the expense of an additional computational time cost. If the molecule crosses the microbeam during this time interval, then the molecule should be reflected from the microbeam's surface at the point of reflection (point I_r at time $t_{i-1} < t < t_i$) to its final position. The reflected molecule undergoes diffusive reflection based on the microbeam's local temperature superimposed with the normal and tangential components of the microbeam's instantaneous velocity at point I_r . Using the DSMC numerical technique, all thermodynamic properties (density, pressure, temperature, etc.) of the gas along with its velocity can be found at any position within the gap film. This way, a complete analysis of the gas behavior along with its effect on the vibrating beam can be performed which sheds more light on the underlying dynamics of the interaction between the moving structure and the surrounding gas.

DIMENSIONLESS GROUPS

To better understand the flow behavior under different conditions, dimensionless groups are defined based on the physical parameters of interest at atmospheric conditions of pressure and temperature. To start with the oscillating micro-cantilever beam, the parameters involved are the beam length (L), beam's tip velocity with amplitude (V_d) and the vibrating frequency (f). For the gas medium surrounding the RF switch, the physical parameters involved are the gas molecular diameter (M_d), the gas molecular mass (M_m), the gas specific heat ratio (γ), the gas density (ρ), the gas mean free path

$$\lambda = \frac{M_m}{\sqrt{2\pi\rho M_d^2}}, \quad (3)$$

the speed of sound in the gas medium

$$a = \sqrt{\frac{\gamma kT}{M_m}}, \quad (4)$$

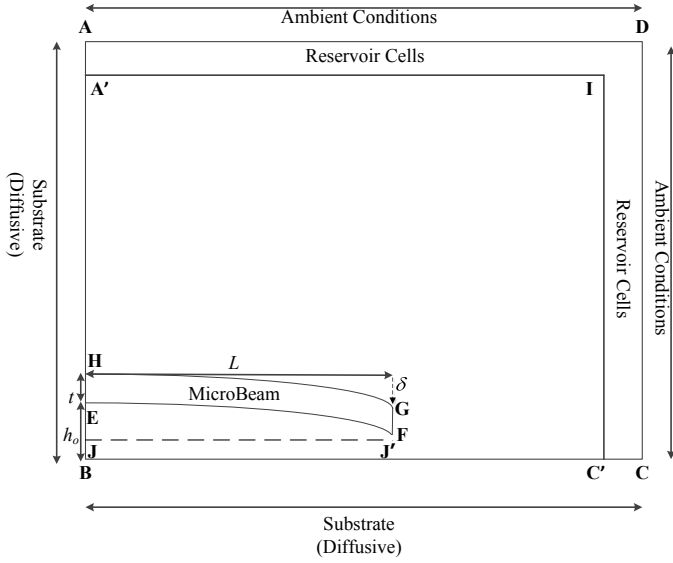


FIGURE 1. TWO DIMENSIONAL SIMULATION DOMAIN.

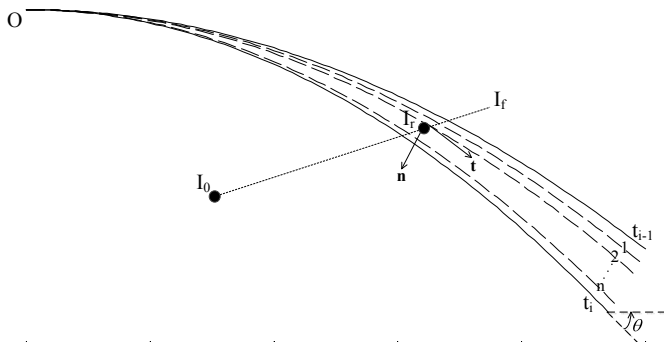


FIGURE 2. DETECTION OF MICROBEAM/MOLECULE COLLISION DURING A TIME STEP.

where k is the Boltzmann constant and T is the temperature, and the gas dynamic viscosity

$$\mu = \frac{M_m}{3\sqrt{2}\pi M_d^2} \sqrt{\frac{8kT}{\pi M_m}}. \quad (5)$$

Since viscosity is a continuum concept and the gas flow in the gap film is in the non-continuum regime ($Kn > 0.01$), the effective viscosity (μ_{eff}) is used according to Veijola's model [23]:

$$\mu_{eff} = \frac{\mu}{1 + 9.638 \left(\frac{\lambda}{h_0}\right)^{1.159}} \quad (6)$$

This correction of the viscosity accounts for the boundary slip flow due to gas rarefaction near the wall.

For the simulation domain, the gap film thickness (h_0) is the main physical parameter influencing the gas flow behavior.

A set of dimensionless groups that capture all the physical parameters mentioned above are formed. The Knudsen number, which determines the degree of rarefaction and the flow regime (continuum vs non-continuum), is the ratio of the molecular mean free path to the characteristic flow length scale

$$Kn = \frac{\lambda}{h_0} \quad (7)$$

The Reynolds number is defined as the ratio of the momentum convection term (inertia forces) to the momentum diffusion term (viscous forces).

$$Re = \frac{V_a h_0}{\nu} = \frac{\tau_{diff}}{\tau_{conv}} \quad (8)$$

where the film gap thickness (h_0) is the characteristic length scale, the beam's tip velocity amplitude (V_a) is the characteristic speed scale, $\nu = \frac{\mu_{eff}}{\rho}$ is the kinematic viscosity, $\tau_{conv} = \frac{h_0}{V_a}$ and $\tau_{diff} = \frac{h_0^2}{\nu}$ are the convection and diffusion time scales respectively. Since the RF switch undergoes harmonic oscillatory motion (unsteady), the Strouhal number (St) is introduced to non-dimensionalize the unsteady component of the inertia term. The Strouhal number is the ratio of inertial forces due to flow oscillations (i.e. local acceleration due to change of velocity with time where the oscillatory time scale is given by $\tau_{osc} = \frac{1}{f}$) to the convection term (i.e. acceleration due to spatial change of velocity)

$$St = \frac{\tau_{conv}}{\tau_{osc}} = \frac{f h_0}{V_a} = \frac{h_0}{2\pi\delta} \quad (9)$$

where (δ), the maximum displacement of the vibrating beam tip, is

$$\delta = \int_0^{\frac{1}{2f}} V_a \sin(2\pi ft) dt = \frac{V_a}{2\pi f} \quad (10)$$

Note that, alternatively, Strouhal number measures the ratio of the gap thickness to the maximum distance traversed by the vibrating beam. Another dimensionless group of interest is the Mach Number (Ma), the ratio of the characteristic gas flow speed to the speed of sound in the gas medium,

$$Ma = \frac{V_a}{a} \quad (11)$$

		Kn	Re (τ_{diff}/τ_{con})	St (τ_{con}/τ_{osc})	Ma	L*	δ/h_0	Re*St (τ_{diff}/τ_{osc})	Ma*Kn (τ_{coll}/τ_{con})	Ma*Kn*St (τ_{coll}/τ_{osc})	Ma*Kn/Re (τ_{coll}/τ_{diff})
	V _a (m/s)	SET 1: Varying V _a (L = 5 μ m, f = 1 GHz)									
	31	0.068	2.927	32.258	0.09	5	0.01	94.43	0.006	0.198	0.002
	157	0.068	14.826	6.369	0.458	5	0.05	94.43	0.031	0.198	0.002
	314	0.068	29.651	3.185	0.915	5	0.1	94.43	0.062	0.198	0.002
	L (μ m)	SET 2: Varying L (f = 1 GHz, V _a = 314 m/s)									
	3	0.068	29.651	3.185	0.915	3	0.1	94.43	0.062	0.198	0.002
	5	0.068	29.651	3.185	0.915	5	0.1	94.43	0.062	0.198	0.002
	10	0.068	29.651	3.185	0.915	10	0.1	94.43	0.062	0.198	0.002
	20	0.068	29.651	3.185	0.915	20	0.1	94.43	0.062	0.198	0.002
f (MHz)	V _a (m/s)	SET 3: Varying f and V _a									
50	78.5	0.068	7.413	0.637	0.229	5	0.5	4.721	0.016	0.010	0.002
100	157	0.068	14.826	0.637	0.458	5	0.5	9.443	0.031	0.020	0.002
200	314	0.068	29.651	0.637	0.915	5	0.5	18.886	0.062	0.040	0.002

TABLE 1. LIST OF DIMENSIONLESS GROUPS FOR THE VARIOUS SIMULATED CASE STUDIES.

where (a), the speed of sound, is expressed as $a = \sqrt{\frac{\gamma k T}{M_m}}$. Finally, the beam length (L) is normalized according to the characteristic length scale (h_0) as

$$L^* = \frac{L}{h_0} \quad (12)$$

Noting that a scales as the molecular most probable speed, the ratios of the collision time scale to convection, diffusion, and oscillation time scales, expressed in terms of dimensionless quantities, are respectively

$$\frac{\tau_{coll}}{\tau_{conv}} \approx \frac{\lambda/a}{h_0/V_a} = (Ma)(Kn) \quad (13)$$

$$\frac{\tau_{coll}}{\tau_{diff}} = \frac{\tau_{coll}}{\tau_{conv}} \frac{\tau_{conv}}{\tau_{diff}} \approx \frac{(Ma)(Kn)}{Re} \quad (14)$$

$$\frac{\tau_{coll}}{\tau_{osc}} = \frac{\tau_{coll}}{\tau_{conv}} \frac{\tau_{conv}}{\tau_{osc}} \approx (Ma)(Kn)(St) \quad (15)$$

RESULTS AND DISCUSSION

Simulations are performed that cover a range of values of the various dimensionless groups as listed in Tab. 1. Table 2 summarizes the geometrical parameters used and the ambient gas medium physical conditions. All results are presented, in terms of normalized quantities, after simulating the flow over a large number of periods to ensure quasi-steady conditions (ie. simulation times are much longer than the transients time that arise from

the initial conditions); the beam's velocity is normalized by its tip velocity amplitude (V_a), the vertical (ie. normal to substrate BC in Fig. 1) and horizontal (ie. parallel to substrate BC) flow field velocities are normalized with respect to the beam's tip velocity amplitude (V_a), pressure is normalized with respect to the relevant gas atmospheric pressure at 20°C, density is normalized with respect to relevant gas density at 20°C, force is normalized with respect to the product of the atmospheric pressure and the microbeam's area, and time is normalized (τ_{norm}) with respect to the beam's oscillation period (T). The relation between the gas flow behavior and the force imparted on the beam is interpreted in terms of the relative importance of diffusion and inertia, where the latter is the superposition of an unsteady term and a convective term. Note that the ratio of the convective inertia term to the diffusion term scales as Re, the ratio of the unsteady inertia term to the convective inertia term scales as St, and the ratio of the unsteady inertia term to the diffusion term scales as the product of Re and St.

Effect of Beam Tip Speed

The first set of simulations investigates the impact of varying the beam speed amplitude (31 - 314 m/s) with beam oscillation frequency of 1GHz. Note that at such high speeds, inertia effects are expected to play a significant role in the gas flow dynamics. The corresponding ranges of Reynolds and Mach numbers are $2.93 \leq Re \leq 29.65$ and $0.09 \leq Ma \leq 0.915$, while ReSt remains constant. Figure 3 shows the dynamic behavior of the gas velocity near the beam, sampled and averaged at the horizontal plane just next to the beam's tip maximum displacement (δ) rep-

Physical Property	Symbol	Value
Beam Length	L	$3\text{ }\mu\text{m}$ to $20\text{ }\mu\text{m}$
Beam Thickness	t	$0.1\text{ }\mu\text{m}$
Initial Gap Height	h_o	$1\text{ }\mu\text{m}$
Excitation Frequency	f	50MHz to 1GHz
Velocity Amplitude	V_a	31 m/s to 314 m/s
Ambient Temperature	T_o	293 K
Ambient Pressure (Air)	P_o	101.1 KPa

TABLE 2. MICRO RF SWITCH GEOMETRY AND GAS FLOW CONDITIONS.

resented by JJ' in Fig. 1. The gas velocity amplitude approaches the vibrating beam's tip velocity amplitude (points B , B' and B'') as Reynolds number increases (note that the microbeam's velocity has been normalized by 10 for all figures merely for scaling purposes). This is expected since larger Reynolds number implies an increasing role of the convective component of the inertia term with respect to diffusion. In terms of molecular behavior, as the beam velocity increases the inertia transfer to molecules (i.e. momentum of convection) becomes more dominant and therefore the gas behavior gets closer to the beam motion governed by τ_{osc} . It is also observed from Fig. 3 that as the beam moves towards the substrate ($0 \leq \tau_{norm} \leq 0.25$), the flow is still moving in the opposite direction away from the substrate due to the inertia effect from the beam motion during the second half of the previous period. It can also be seen that during the first half period when the beam is moving towards the substrate, the flow near the beam reverses the direction of its motion at approximately the same time for all the cases considered. At $\tau = 0.25$, the beam reaches maximum speed half way through the gap, and for $\tau > 0.25$, it starts slowing down and the flow in its vicinity reverses its direction and starts moving in the same direction as the beam. At $\tau = 0.5$ when the beam has undergone maximum displacement and the gap size is minimum, the flow in its vicinity, with inertia that is dependent on V_a (and thus Re) is still moving towards the substrate. At $\tau > 0.5$, the beam starts moving away from the substrate, while the flow in its vicinity is observed to still move towards the substrate. At this point, we observe different behaviors among the different cases. By inspecting points C , D and E in Fig. 3, it can be observed that for $0.5 < \tau < 0.75$, the flow in the beam vicinity starts slowing down for the case with $Re = 29.651$ while it continues to move towards the substrate for the case with $Re = 2.927$. This can again be attributed to inertia effects; the flow inertia is related to beam inertia in the sense that for larger beam speeds, it imparts larger momentum on the nearby molecules. So as the beam starts moving away from the substrate, how quickly the flow reverses its direction and follows it depends on both the beam speed and the Reynolds number of the flow. For small V_a , the beam slow motion allows the flow

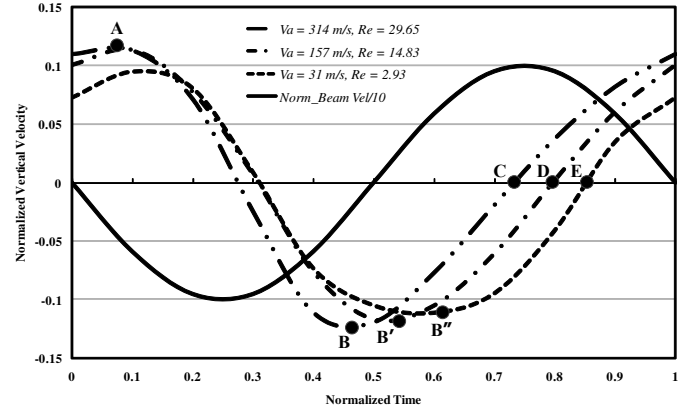


FIGURE 3. AVERAGE VERTICAL FLOW VELOCITY OF GAS FILM AT PLANE NEXT TO THE BEAM'S MAXIMUM TIP DISPLACEMENT WITHIN THE GAS FILM GAP FOR DIFFERENT BEAM TIP SPEEDS.

(with smaller Re) to take longer time to responds to the beam change in motion direction, whereas for larger V_a , the fast moving beam (away from the substrate) forces the flow to respond more quickly. In summary, the cases with larger Re have asymmetric response of the flow velocity in its vicinity due to strong inertia effects, whereas the response is close to harmonic for the small Re case. Moreover, Fig. 3 shows that the flow reverses its direction during the first half period (rise time from A to B) much faster than it does in the second half period (drop time from B back to A) for flows with high flow inertia (ie. Re). This is due to the fact that during the first half period, the beam is moving towards the substrate while the flow with high Re is still moving in the opposite direction towards the beam itself. This forces the flow to respond quickly to the beam motion as compared to the second half period where the beam moves away from the substrate while the flow inertia propagates more freely towards the substrate without hitting the beam itself.

In terms of the beam's front (BF) face force (solid line), the same inertia effect explained before imposes a high force amplitude during the first half period (collision of a relatively high number of molecules moving away from substrate due to inertia effects with the beam moving towards the substrate) as compared to the second half period (molecules keep moving towards the substrate due to inertia effects while the beam is moving away from the substrate). The same phenomena can be observed on the beam's back (BB) face force (dashed line) with the difference that during the second half period the flow propagates away from the beam towards the ambient boundary rather than the substrate. Figure 4 also shows that the force amplitude on the beam's front face (facing the substrate) during the first half of the period (as the beam moves towards the substrate) is larger than the force amplitude on the beam's back face during the second half of the

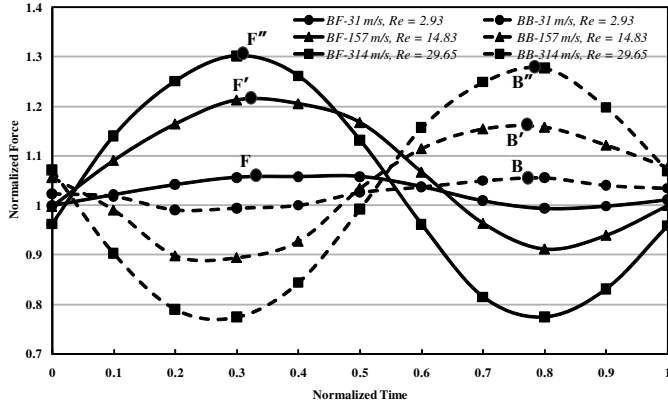


FIGURE 4. FORCES AT BEAM'S FRONT AND BACK FACES FOR DIFFERENT BEAM TIP SPEEDS.

period (as the beam moves away from the substrate). This is expected due to the squeeze film effect; the back face is open to the atmosphere whereas the front face senses the higher pressure of the squeeze film in the gap. The amplitude of the net force on the beam (difference between force on front face and force on back face) increases for larger values of Re . As can be seen from Fig. 4, as Re increases, the beam's front face force amplitude increases as seen by points F , F' and F'' . Same trend is observed at the beam's back face force (dashed line) amplitude represented by points B , B' and B'' . This behavior is expected since convection plays a bigger role as Re increases, which is associated with larger momentum transfer to the beam due to collisions with fluid molecules, thus exerting a larger force on the vibrating beam. Further inspection of the force acting on the beam front face (Fig. 4) shows that when the oscillating beam stops at half the period of motion (i.e. at $\tau_{norm} = 0.5$), the force at the beam's front face is larger than that exerted by atmospheric pressure and this value increases with increasing Re (ie. higher film squeezing effects) whereas at the beam's back face the force is much closer to that of atmospheric pressure (no squeeze effect).

Effect of Micro-Cantilever Beam's Length

In the second set of simulations, the beam's tip speed amplitude is fixed at 314 m/s and the frequency at 1 GHz while the microbeam's normalized length L^* is varied from 3 to 20. Figures 5 and 6 show respectively time evolution of the average gas velocity (at plane JJ' in Fig. 1) and the forces acting on the beam back and front beam face forces. Since Reynolds and Strouhal numbers are constant, the relative contribution of diffusion with respect to convection (Re), convection with respect to the beam's oscillatory motion (St), and diffusion with respect to the beam's oscillatory motion ($ReSt$) do not change. Therefore, the gas flow velocity for different microbeam's lengths is expected to be the same as depicted in Fig. 5 where the inertia and diffusion effects

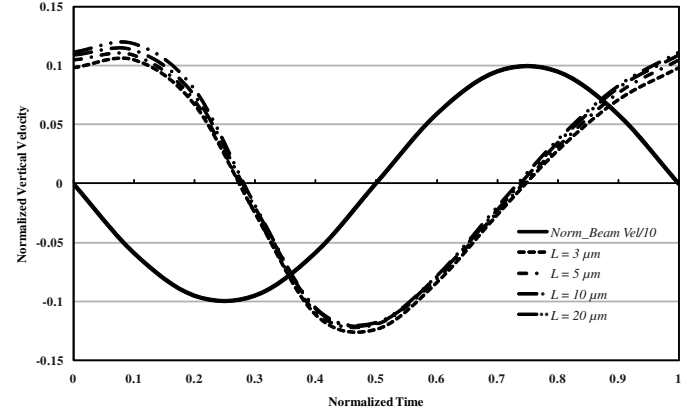


FIGURE 5. AVERAGE VERTICAL FLOW VELOCITY OF GAS FILM AT PLANE NEXT TO THE BEAM'S MAXIMUM TIP DISPLACEMENT WITHIN THE GAS FILM GAP FOR DIFFERENT BEAM LENGTHS.

have equal contributions on the flow velocity profile.

The forces exerted by the gas on the oscillating beam front and back faces are shown in Fig. 6. The force amplitude on the beam's front face during the first half of the period (as the beam moves towards the substrate) is larger than the force amplitude on the beam's back face during the second half of the period (as the beam moves away from the substrate) due to the squeeze film effect as explained for the first simulation case. However, it is observed that the force on the beam's front face is independent of the micro-cantilever beam length for values of the beam length that are much larger than the film gap thickness (ie. $L^* \geq 10$). As the microbeam's length gets smaller with respect to the film gap thickness, the front face force increases, as depicted on Fig. 6, and this is mainly due to the deflection profile of the beam where the microbeam's tip deflection angle (θ on Fig. 2) increases with decreasing microbeam's lengths to keep a constant microbeam tip displacement. Figure 7 shows the average horizontal (parallel to substrate BC on Fig. 1) flow field velocity that has a positive value (ie. flow away from substrate AB) for the relatively long microbeam (20 μm) and starts inverting it's direction till it reaches a negative direction (ie. flow towards substrate AB) for the short microbeam (1 μm) with relatively large deflection tip angle that tends to force the flow towards the inside region of the film gap (EB on Fig. 1) and thus increases the squeezing effects. The results of Fig. 6 are in good agreement with the analytical work of Hosaka et al. [15] where they showed that both squeezed and free space damping coefficients decrease with increasing beam length (the rate of decrease for the former represented by the front beam force is higher than that for the latter represented by the back beam force) under the limitations of applicability of the Reynolds equation (ie. inertia free, very small tip amplitude displacement, etc.).

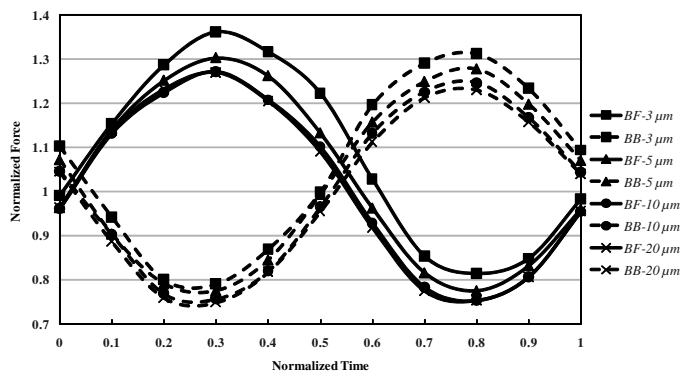


FIGURE 6. FORCES AT BEAM'S FRONT AND BACK FACES FOR DIFFERENT BEAM LENGTHS.

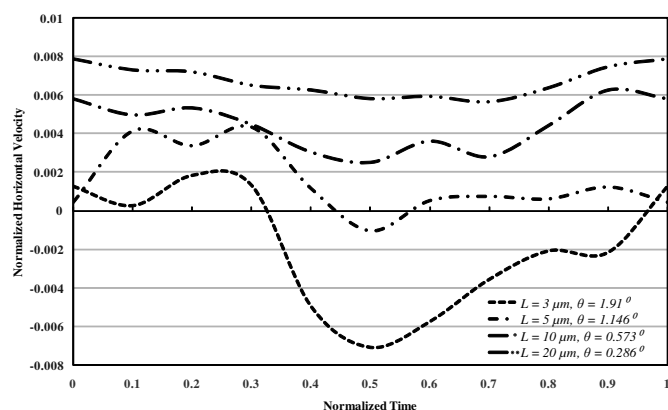


FIGURE 7. AVERAGE HORIZONTAL FLOW VELOCITY OF GAS FILM AT PLANE NEXT TO THE BEAM'S MAXIMUM TIP DISPLACEMENT WITHIN THE GAS FILM GAP FOR DIFFERENT BEAM LENGTHS.

Effect of Reynolds Number (With Constant Strouhal Number)

For the third set of simulations, the beam's tip velocity amplitude V_a is varied from 78.5 m/s to 314 m/s and the frequency f is accordingly varied from 50 MHz to 200 MHz to maintain a fixed $St = 0.6366$ where the beam's tip traverses half the film gap thickness. Since Strouhal is constant, the effect of the convective inertia term with respect to the unsteady oscillatory term is constant, and hence the decisive term that governs the phase difference between the beam oscillatory motion and the gas flow motion is the ratio of inertia to diffusion (ie. Re and $ReSt$). As Re decreases, the convective inertia term effect decreases (ie. the gas flow velocity magnitude decreases as compared to that of the microbeam as seen by points B , B' and B'' in Fig. 8) and simultaneously the product of Re and St decreases forcing the flow velocity to get closer in phase to the vibrating beam motion (dif-

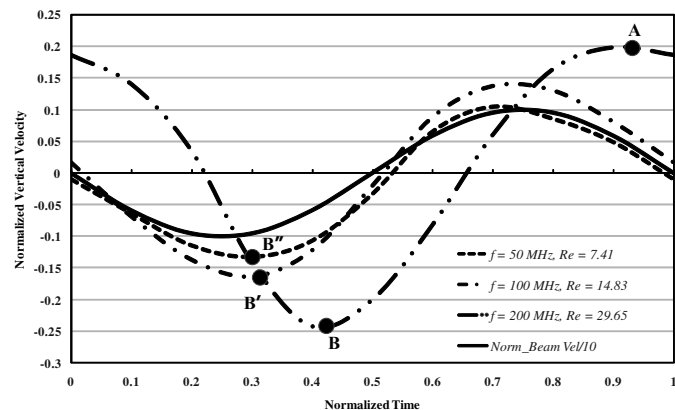


FIGURE 8. AVERAGE VERTICAL FLOW VELOCITY OF GAS FILM AT PLANE NEXT TO THE BEAM'S MAXIMUM TIP DISPLACEMENT WITHIN THE GAS FILM GAP FOR DIFFERENT OSCILLATING FREQUENCIES AND BEAM TIP SPEEDS.

fusion time scale approaches microbeam oscillatory time scale). As compared to the first simulation set, the gas flows for this simulation set have almost the same rise and drop times with no deviation from the sinusoidal behavior. This could be explained by comparing the flow behavior in the first simulation set for the beam's tip velocity of 314 m/s at 1 GHz with the flow behavior for the current simulation set at a beam's tip velocity of 314 m/s at 200 MHz frequency. For both cases during the first half period, the flow reaches its maximum velocity propagating towards the fixed substrate (ie. same direction of motion as beam) at almost the same normalized time as depicted by point B in Fig.'s 3 and 8. When the beam inverts its oscillation direction allowing the flow to propagate away from the beam, hit the fixed substrate, and reflect back to hit the oscillating beam. Since the microbeam for the first simulation set has a very high frequency of 1 GHz (ie. very fast oscillating time), the reflected flow hits the beam after it has completed its period and started a new cycle moving back towards the substrate as seen by point A on Fig. 3 whereas for the much slower oscillating beam (200 MHz), the reflected flow will hit the beam while it's still moving away from the substrate during the same oscillating cycle as depicted by point A in Fig. 8 leading to insignificant difference between the rise and drop times. Therefore, asymmetric behavior of the gas flow emerges when the beam is moving at high speeds (ie. high inertia effects) for high oscillating frequencies.

The same inertia effect explained in the first simulation set shows a high force amplitude during the first half period relative to the second half period for each of the beam's front face force (solid line) and back face force (dashed line) for high inertia flows as depicted on Fig. 9. One interesting behavior is depicted for the relatively low oscillating frequency (ie. 50 MHz) where the difference between the front and back beam faces force

during the first half period (ie. $0 \leq \tau_{norm} < 0.5$) is much larger than that during the second half period (ie. $0.5 < \tau_{norm} \leq 1$). To explain this behavior, the flow field velocity at half period (ie. $\tau_{norm} = 0.5$) for the microbeam oscillating at 50 MHz is examined in Fig. 10 where the x and y axes represent the simulation domain in μm . Circulation effects are caused by the highly squeezed gas film beneath the front beam face which escapes from the microbeam's boundary, rotates towards the low pressure region at the beam back face and hits it thus imposing a relatively higher pressure than expected. It's worth noting that this circulation phenomenon shows up at this relatively low oscillating frequency simulation since the microbeam's tip is moving through half the film gap and thus squeezing the air film outside the gap (ie. no gas "entrapment" since there is "enough" time for the gas to escape the gap region due to the long oscillation period). This circulation effect also causes the beam back face force profile to follow the front face force profile where both peak at almost half the period. Another interesting behavior is depicted at the front face force where it follows the beam velocity profile (ie. maximum at $\tau_{norm} = 0.25$ and minimum at $\tau_{norm} = 0.75$) at high frequency (200 MHz) and the displacement profile (ie. maximum at $\tau_{norm} = 0.5$ and minimum at $\tau_{norm} = 0$ and 1) for the lower frequency simulations (50 and 100 MHz). To closely study such behavior and examine this transition of front face force profile, a series of additional simulations are performed for a set of frequencies that shows in Fig. 11 the gradual change in the profile of the force acting on the beam front face. This force combines the dissipative and compressibility effects of the squeezed gas film and may therefore be modeled according to the equation

$$F(t) = Ky_{tip} + C\dot{y}_{tip} + \phi \quad (16)$$

where K represents the compressibility factor, C represents the dissipative factor, and ϕ is the inertia effect.

Table 3 shows the values of K , C and ϕ for the simulated cases in Fig. 11 where compressibility effects dominate at the relatively low frequencies (this means that the current simulation frequencies are already above cutoff otherwise dissipative effects would have dominated). As the frequency increases further (much higher than cutoff), the dissipative behavior of the film gap dominates over the compressibility effects since at very high frequencies the beam is oscillating so fast that there is not "enough" time for the gas film gap to get compressed between the microbeam and the substrate. In terms of molecular behavior and for compressibility effects to show, the molecules reflecting from the microbeam should have enough time to collide with their neighboring molecules and those in turn with the molecules around them till these effects reach the substrate underneath and reflect back to the beam. Table 3 also shows that at very high frequencies, the "negative" spring coefficients reflect the domi-

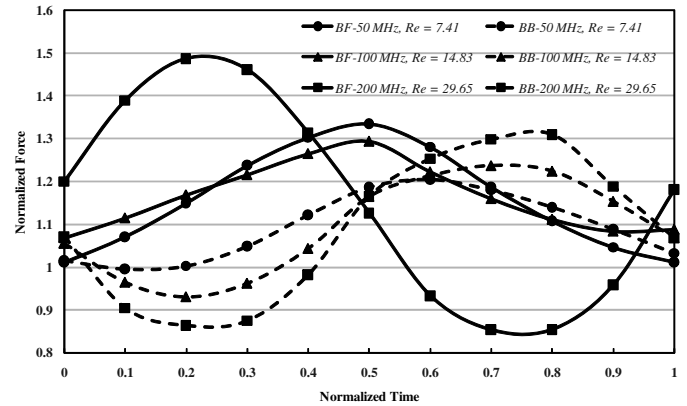


FIGURE 9. FORCES AT BEAM'S FRONT AND BACK FACES FOR DIFFERENT OSCILLATING FREQUENCIES AND BEAM TIP SPEEDS.

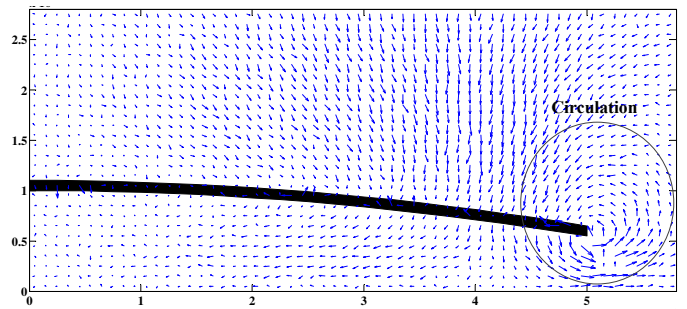


FIGURE 10. FLOW VELOCITY FIELD OF GAS FILM IN THE VICINITY OF THE OSCILLATING MICROBEAM AT $\tau_{norm} = 0.5$.

nance of inertia forces (increasing Re) as discussed by Veijola et al. [9] for a capacitive RF switch and therefore both negative K and ϕ contribute to inertia effects. If inertia effects were negligible, the value of ϕ which indicates the normalized front face force at $\tau_{norm} = 0$ or 1 where the beam's tip velocity and displacement are both zero, should be unity.

CONCLUSION

The dynamic force on a micro-cantilever beam oscillating above a substrate imparted by a surrounding medium is investigated for a range of oscillating speed amplitudes, frequencies, and microbeam's length. The micro flow behavior of the gas film between the beam and the substrate and its relation to the forces acting on the front and back sides of the beam are interpreted in terms of the relative contributions of diffusion and the unsteady and convective components of inertia, as quantified by the various dimensionless groups. It has been observed that for larger values of Reynolds number, the force on the side of the

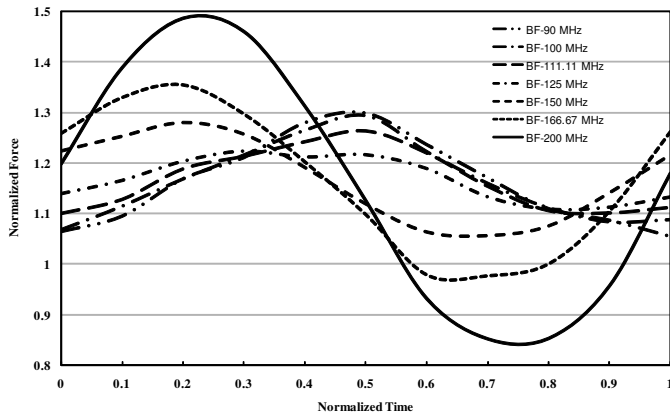


FIGURE 11. FORCES AT BEAM'S FRONT FACE FOR DIFFERENT OSCILLATING FREQUENCIES.

f (MHz)	K	C	ϕ
50	0.304	0.023	1.019
90	0.217	0.025	1.063
100	0.19	0.03	1.076
111.11	0.145	0.033	1.1
125	0.076	0.045	1.133
150	-0.093	0.106	1.213
166.67	-0.163	0.18	1.243
200	-0.062	0.331	1.187

TABLE 3. LIST OF DISSIPATIVE, COMPRESSIBILITY AND INERTIA COEFFICIENTS FOR DIFFERENT SIMULATION FREQUENCIES.

beam facing the substrate (front face) departs from the harmonic profile of the beam motion imposed, the flow velocity rise time decreases with respect to drop time, the force amplitude at beam front face during the first half period (i.e. when beam is moving towards substrate) is larger than the beam back face force during the second half period (beam moving away from substrate) due to squeeze damping effects, and the force amplitude on both beam faces during the first half period is larger than their counterparts during the second half period due to inertia effects. For small Strouhal number, the phase difference between the beam and flow within the gap decreases. When Re and St are small, diffusion plays a dominant role over inertia, and consequently drives the flow velocity closer to the beam's motion while decreasing the front face force amplitude. This corresponds to relatively low squeeze number where the gas has enough time to diffuse and escape from getting trapped between the oscillating beam and the fixed substrate leading to circulation effects at the micro-cantilever beam's tip. By decreasing the length of the micro-cantilever beam, the squeeze and free film damping forces

on the beam's front and back faces increase with the former increasing at a higher rate.

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